

Mathematics A

Unit 1 • Chapter OL-T31 Classified

Cross-session • chapter-organised candidate

10 classified items • 43 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Unit 1 • Chapter OL-T31 • 10 items • 43 marks

Chapter	Items	Marks	Starts
01. OL-T31 Coordinate Geometry	10	43	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Coordinate Geometry

Unit 1 • 43 marks • 10 item(s)

June 2023 | 4MA1-2H Q24 | 6 marks

Kite or symmetric vertex coordinates

June 2022 | 4MA1-2H Q22 | 7 marks

Kite or symmetric vertex coordinates

January 2020 | 4MA1-1H Q1 | 2 marks

Midpoint from two endpoints

January 2020 | 4MA1-1H Q22 | 6 marks

Coordinates from isosceles and gradient constraints

January 2021 | 4MA1-2H Q18 | 4 marks

Rectangle or parallelogram missing vertex

May-Nov 2020 | 1H Q22 | 4 marks

Read a plotted coordinate pair

January 2023 | 1H Q23 | 6 marks

Read a plotted coordinate pair

January 2023 | 2H Q6 | 3 marks

Rectangle or parallelogram missing vertex

November 2025 | 1H Q20 | 3 marks

Read a plotted coordinate pair

November 2025 | 2H Q1 | 2 marks

Rectangle or parallelogram missing vertex

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Question 28

4MA1-2H Q24

MODULAR UNIT 1

6 marks

24 $ABCD$ is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

(..... ,)

WORKING SPACE

Question 28

4MA1-2H Q22 Q22

MODULAR UNIT 1

7 marks

- 22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D .

(..... ,)

(..... ,)

WORKING SPACE

- 1 The point A has coordinates $(5, -4)$
The point B has coordinates $(13, 1)$

(a) Work out the coordinates of the midpoint of AB .

(.....,)
(2)

YOUR WORKING

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$

J is the point with coordinates $(j, 15)$ where $j < 0$

K is the point with coordinates $(6, k)$

M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

YOUR WORKING

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

YOUR WORKING

22 $ABCD$ is a rhombus.

The diagonals, AC and BD , intersect at the point M .

The coordinates of M are $(6, -11)$

The points A and C both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through B and D intersects the y -axis.

(.....,)

(Total for Question 22 is 4 marks)

YOUR WORKING

23 $ABCD$ is a kite.

$$AB = AD \text{ and } CB = CD$$

The point B has coordinates $(k, 1)$ where k is a negative constant.

The point D has coordinates $(8, 7)$

The straight line L passes through the points B and D

The straight line L is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC

Give your answer in the form $px + qy = r$ where p , q and r are integers.

Show your working clearly.



YOUR WORKING

(Total for Question 23 is 6 marks)

YOUR WORKING

6 The points A and B are on a coordinate grid.

The coordinates of A are $(6, 4)$

The coordinates of B are $(17, j)$ where j is a constant.

The midpoint of AB has coordinates $(k, 15)$ where k is a constant.

Find the value of j and the value of k

$j =$

$k =$

(Total for Question 6 is 3 marks)

YOUR WORKING

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**

Give your answer in the form $y = mx + c$

(Total for Question 20 is 3 marks)

YOUR WORKING

- 1 The point P has coordinates $(3, 4)$
The point Q has coordinates $(9, 16)$

M is the midpoint of the line PQ

Find the coordinates of M

(..... ,)

(Total for Question 1 is 2 marks)

YOUR WORKING
